

DS	Session	Searching	Question Information	Level 1	Challenge 10
Question Description: An array A contains integers with the following constraints: 'A' contains elements in sorted order. Integer i occurs $i \cdot \text{floor}(\sqrt{i}) + \text{ceil}(i/2)$ times in the array. All elements are greater than or equal to 1. Fathima has given Q queries of type: L R: Find the number of distinct values present in subarray A[L...R]. Note: 1-based indexing is followed. Constraints: $1 \leq Q \leq 10^5$ $1 \leq L \leq R \leq 10^{13}$ Input Format: The first line contains an integer Q denoting the number of queries. Next Q lines contains two space-separated integers L, R, denoting the query. Output Format: Print the output in separate lines contains required number of distinct values. Example Sample Input: 2 1 3 1 6 Sample Output: 2					

Explanation

First few elements of the array A are 1, 1, 2, 2, 2, 3, 3, 3, 3, 3, ...

- For Query 1:-
 - Number of distinct elements in subarray $A[1...3]$ is 2.
- For Query 2:-
 - Number of distinct elements in subarray $A[1...6]$ is 3.

Logical Test Cases

Test Case 1

INPUT (STDIN)

2
3 3
2 6

EXPECTED OUTPUT

1
3

Test Case 2

INPUT (STDIN)

3
3 3
2 6
5 4

EXPECTED OUTPUT

1
3
1

Mandatory Test Cases

Test Case 1

KEYWORD

while(l<ans1)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

400

Test Case 3

NLOC

56